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Marks	0/14
Grade	0 out of 100

Question 1

Not answered

Marked out of 1

An environmental agency identifies 200 patients with a lifetime history of panic attacks and 200 matched controls without the condition. Exposure to loud traffic noises was compared in two groups. 20 patients and ten controls had such an exposure. Which of the following is false with respect to the above study design?

Select one:

- ☐ Both exposure and disease have occurred before the onset of the study.
- ☐ It is less cost effective than following up those who were exposed to loud noise till they have panic attacks
- ☐ It is mostly useful to study outcomes which are rare
- ☐ It is mostly useful in diseases where exposure is common
- ☐ It is a retrospective design

Your answer is incorrect.

The above is an example of case-control study design. The exposure and the outcome have both taken place when the subjects are recruited. It is retrospective, cheap compared to cohort studies and is useful to study common exposures (but relatively rare outcomes).

The correct answer is: It is less cost effective than following up those who were exposed to loud noise till they have panic attacks

Question 2

Not answered

Marked out of 1

In a case-control study of possible nutritional correlates of depression, subjects were asked about the frequency of consuming 20 different food items in a Likert scale. The histories of cases differed significantly from those of controls, at $p=.05$, for three of the 20 listed items. Which of the following is most likely?

Select one:

- ☐ The study is inconclusive: further case control studies with more rigorous methods are required
- ☐ Diet is related to depression; but it is unclear which of the three items is the most important culprit
- ☐ Diet is not related to depression as the results are due to chance finding: discard the study
- ☐ The results are conclusive if it could be shown that there was no recall bias
- ☐ Diet is definitely related to depression; further cohort studies for each of the 3 items are warranted

Your answer is incorrect.

There are two important problems with this study. Asking depressed subjects to recall numerous food items can produce serious recall bias. In addition, at a significance level of 0.05, at least one out of the 20 items may show an association by chance. As it is difficult to say which of the three items are really associated with depression, it may not be wise to invest huge effort and money in a cohort study at this stage. It is best if a more rigorous case-control study is attempted to replicate or refute the findings.

The correct answer is: The study is inconclusive: further case control studies with more rigorous methods are required

Question 3

Not answered

Marked out of 1

A case-control study investigating various risk factors for a disease in hospitalised patients chooses healthy relatives of the patients as controls. In which of the following situations the results may be distorted?

Select one:

- ☐ Risk factor is related to diet and lifestyle
- ☐ Risk factor is purely related to occupational exposure
- ☐ Disease is not shared by families
- ☐ Exposure is likely to be not shared by families
- ☐ Risk factor is not related to household environment

Your answer is incorrect.

This matching approach increases the likelihood that the factors that are generally shared by family members will be similar for both the controls and the cases. This reduces bias and confounding especially if the cause of a disease is not shared in families. Unfortunately in cases where the causal agent is genetic or otherwise familial, this method leads to "overmatching" thus reducing the ability to demonstrate the causation.

The correct answer is: Risk factor is related to diet and lifestyle

Question 4

Not answered

Marked out of 1

Which one of the following types of bias is NOT a major concern in a hospital-based case-control study?

Select one:

- ☐ Verification bias
- ☐ Berkson bias
- ☐ Neyman bias
- ☐ Recall bias
- ☐ Family information bias

Your answer is incorrect.

Work up bias (verification bias) refers to a systematic error in the assessment of the validity of a diagnostic test. When the execution of the gold standard is influenced by the results of the assessed test, especially when the reference test is less frequently performed when the assessed test result is negative, then this will influence the number of false negatives correctly identified in the exercise. This bias is specific to assessment of diagnostic tests and so not seen in ordinary case-control studies where cases and controls are determined before estimates of exposure begin. Family aggregation bias is a type of information bias wherein presence of a case in the family increases awareness of risk factors and exposure leading to more family members of cases reporting an exposure though controls have family members with less awareness. Journal Epid Comm Health 2004;58;635-641

The correct answer is: Verification bias

Question 5

Not answered

Marked out of 1

Which of the following describes the relationship between prevalence and incidence rates?

Select one:

- ☐ Prevalence = incidence/duration
- ☐ Prevalence = Incidence X duration
- ☐ Incidence = prevalence X duration
- ☐ Prevalence = incidence - duration
- ☐ Prevalence X incidence = duration

Your answer is incorrect.

In a steady-state situation, in which the rates are not changing, and in-migration equals out-migration, the following equation applies - Prevalence = Incidence × Duration of disease. Gordis, L. Epidemiology 4th ed. Chapter 3 Saunders Elsevier

The correct answer is: Prevalence = Incidence X duration

Question 6

Not answered

Marked out of 1

In a study examining the relationship between domestic abuse and depression, women who do and do not suffer abuse are followed up over a five-year period. It is found that 70 of the 500 individuals who undergo domestic abuse develop depression; 150 of 3000 individuals who don't get abused develop depression. If the prevalence of domestic abuse in a particular council is 20%, the expected incidence of depression in this council is

Select one:

- ☐ 0.068
- ☐ 5.2
- ☐ 1.6
- ☐ 0.02
- ☐ 3.44

Your answer is incorrect.

It = (Prevalence in the exposed)(Incidence in the exposed) + (1-Prevalence in the exposed)(Incidence in the non-exposed) = (20%)(70/500) + (1-20%)(150/3000) = 0.068 or 6.8%

The correct answer is: 0.068

Question 7

Not answered

Marked out of 1

Cohort studies allow researchers to calculate relative risk and attributable risk. Which of the following statements regarding these measures is true?

Select one:

- ☐ Attributable risk is a cross product
- ☐ Attributable risk cannot take negative values
- ☐ Relative risk cannot take negative values
- ☐ Attributable risk but not relative risk explains the reason for differences between groups
- ☐ Attributable risk cannot be zero

Your answer is incorrect.

Relative risk is a ratio - it can have values less than 1, but not less than 0. Attributable risk is an absolute risk difference it can be less than zero when the risk in exposed is less than the risk in non-exposed. Both attributable risk and relative risk are measures of differences between groups - while the former is an absolute difference, the latter is a ratio. The odds ratio is a cross product.

The correct answer is: Relative risk cannot take negative values

Question 8

Not answered

Marked out of 1

National Comorbidity Survey examined a possible association between traits of neuroticism and various psychiatric disorders. Trained non-clinical staff screened randomly chosen individuals using well-validated questionnaires to establish lifetime diagnoses in addition to collecting data on drug use, alcohol intake and various personality traits. Which of the following measure of disease occurrence can be calculated from this study?

Select one:

- ☐ Risk rate
- ☐ Incidence density
- ☐ Relative risk
- ☐ Incidence rate
- ☐ Prevalence

Your answer is incorrect.

Cross-sectional surveys can estimate prevalence only; to find out incidence rates a cohort study or repeated cross-sectional surveys at two time points is necessary.

The correct answer is: Prevalence

Question 9

Not answered

Marked out of 1

A study looks into patient records for a group of schizophrenic patients admitted to a psychiatric hospital between 1934 and 1944. All of the patients were deceased at the time of study. Each patient's social history recorded at the time of admission was scored by two independent raters for the quality of social resources. The impact of social support on the longevity of schizophrenia patients was examined. This study can be described as

Select one:

- ☐ Controlled trial
- ☐ Cross sectional study
- ☐ Ecological study
- ☐ Retrospective cohort study
- ☐ Prospective cohort study

Your answer is incorrect.

This study involved retrospective data collection. The groups were separated according to the level of social input at the time of admission (exposure) and then the longevity of life (outcome - which occurred later) was examined. This is a retrospective cohort design.

[/www.psychosomaticmedicine.org/cgi/content/full/61/2/146](http://www.psychosomaticmedicine.org/cgi/content/full/61/2/146)

The correct answer is: Retrospective cohort study

Question 10

Not answered

Marked out of 1

Which of the following study designs is useful to measure point prevalence of a clinical problem?

Select one:

- ☐ Cross sectional study
- ☐ Case control study
- ☐ Cohort study
- ☐ Controlled trial
- ☐ Uncontrolled trial

Your answer is incorrect.

A cross-section survey assesses the status of an individual with respect to the presence or absence of a disease at a given point in time. It can be used to assess point prevalence.

The correct answer is: Cross sectional study

Question 11

Not answered

Marked out of 1

Which of the following study designs is useful to measure the incidence of a clinical problem?

Select one:

- ☐ Uncontrolled trial
- ☐ Controlled trial
- ☐ Single cross sectional study
- ☐ Case control study
- ☐ Cohort study

Your answer is incorrect.

Incidence is defined by number of new cases. One needs to observe a population long enough to pick all new cases. This is achieved by cohort studies.

The correct answer is: Cohort study

Question 12

Not answered

Marked out of 1

Which of the following is an advantage of case-control design over cohort design?

Select one:

- ☐ Ability to calculate incidence rates
- ☐ Can be used to determine survival rates
- ☐ Absence of recall bias
- ☐ Can be used to study prodromal course of a disease
- ☐ Can explore multiple risk factors for the same disease

Your answer is incorrect.

Case-control studies are cheaper and faster, compared with other analytic designs. They are useful for studying diseases with long latent periods. They are useful for studying rare diseases since subjects are selected on the basis of disease status. They allow the investigation of multiple exposures simultaneously

The correct answer is: Can explore multiple risk factors for the same disease

Question 13

Not answered

Marked out of 1

Which of the following study design is suitable for survival analysis?

Select one:

- ☐ N of 1 trials
- ☐ Cross over trials
- ☐ Cross sectional epidemiological surveys
- ☐ Cohort studies
- ☐ Case control studies

Your answer is incorrect.

Survival data measure follow-up time from a defined starting point to the occurrence of a given event, for example the time from the beginning to the end of a remission period or the time from the diagnosis of a disease to death. Such observations are made possible only by studies using follow-up data (mostly prospective, sometimes retrospective) such as cohort studies. It is also possible in parallel RCT designs but usually not attempted in cross over studies where the observation periods are fairly short (retrieved from <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC1065034/>).

The correct answer is: Cohort studies

Question 14

Not answered

Marked out of 1

A research team wanted to study the association between an occupational exposure and a relatively rare disease. After much struggle, they obtain access to the employment records of several major establishments. From these records, the exposures of workers over a period of 10 years are obtained. Using national hospital registry and public health observatory data the researchers determine the cumulative incidence of disease after exposure in factory workers divided into high, low and no exposure. What type of study is this?

Select one:

- ☐ Cross sectional survey
- ☐ Retrospective cohort study
- ☐ Audit
- ☐ Controlled trial
- ☐ Case control study

Your answer is incorrect.

As incidence rates are calculated, this study is a cohort study. The subjects are ascertained by exposure status and then the outcome rates are studied using registries. So this is a retrospective cohort study.

The correct answer is: Retrospective cohort study



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